

ASPIRE COLLEGE VEHARI CAMPUS

Notes • BS CS 3rd Semester • Lecture 13: Connecting Devices & Network Bridging

Course: Computer Networks (Credit Hours: 3)

Instructor: Muhammad Usman Ghani

1. WHAT IS INTER-NETWORKING & CONNECTING DEVICES?

To expand networks and allow computers on different cables or buildings to talk, we use specialized networking hardware. These devices operate at different layers of the network architecture.

Layer 1: Hub & Repeater

- **Repeater:** Regenerates weak, attenuated electrical signals over long distances.
- **Hub (Multiport Repeater):** Dumb device. When it receives a frame, it blindly broadcasts (floods) it to *all connected ports*. Creates huge collision domains.

Layer 2: Bridge & Switch

- **Bridge:** Connects two separate LAN segments; filters traffic based on MAC addresses.
- **Switch (Multiport Bridge):** Intelligent device. Reads Destination MAC address and forwards frames *only to the intended port* (micro-segmentation).

2. NETWORK BRIDGING (HOW A BRIDGE WORKS)

A **Bridge** connects two or more LAN segments (e.g., Lab A and Lab B) and filters frames using a **MAC Address Table (Forwarding Database)**.

How Bridges Learn (Transparent Bridging):

- **Backward Learning:** When a frame enters port 1 from PC-A, the bridge notes down: "*PC-A's MAC is on Port 1*".
- **Filtering:** If PC-A sends data to PC-B (both in Lab A), the bridge blocks the frame from entering Lab B, saving bandwidth.
- **Forwarding / Flooding:** If destination MAC is unknown, it floods the frame to all other ports until the device responds.

3. COLLISION DOMAINS VS. BROADCAST DOMAINS

Understanding how networking devices control traffic flow and congestion:

Concept	Definition & Behavior	Which Device Breaks It?
Collision Domain	A network segment where data packets can collide with each other if sent at the same time.	Bridges & Switches break collision domains (each switch port is its own collision domain).
Broadcast Domain	The entire network area where a broadcast frame (sent to FF:FF:FF:FF:FF:FF) is heard by every computer.	Routers (Layer 3) break broadcast domains (Switches forward broadcasts everywhere).

4. SUMMARY COMPARISON: HUB VS. BRIDGE VS. SWITCH VS. ROUTER

Device	Operating Layer	Addressing Used	Forwarding Method
Hub	Layer 1 (Physical)	None (does not read data)	Broadcasts to all ports indiscriminately
Bridge	Layer 2 (Data Link)	MAC Address (Software-based)	Filters and forwards between 2-4 segments
Switch	Layer 2 (Data Link)	MAC Address (Hardware ASIC chip)	Forwards directly to specific destination port
Router	Layer 3 (Network)	IP Address (Logical)	Routes packets across different networks / WANs

5. QUICK STUDENT REVIEW QUESTIONS

Q1: Why is a Switch called an "Intelligent Hub"?

Ans: Because a Hub blindly repeats data to all ports causing collisions, while a Switch reads MAC addresses and sends frames only to the target device's specific cable port.

Q2: If a Switch has 8 ports, how many collision domains and broadcast domains does it have?

Ans: It has **8 Collision Domains** (1 per port) and **1 Broadcast Domain** (broadcasts flood to all 8 ports).